

Sequencing at an Inflection

Demand signals from the operator front line — NGS platforms & liquid-biopsy screening

EXECUTIVE SUMMARY

The next-generation sequencing supply market is destabilising faster than incumbent-moat models imply. Drawing on the ReaDepth panel of ~360 sequencing & diagnostics contributors, we find latent demand for Roche's new entrant running well ahead of published expectations, a switching moat thinner than commonly assumed, challenger enthusiasm concentrated among industry insiders rather than buyers, and clinical assay demand concentrating by class rather than uniformly.

Roche/Axelios demand exceeds the published view. Contributors break nearly 30 to 1 in Roche's favor — for every one who sees its momentum fading, roughly 30 see it rising — and 45% of clinical test vendors are already considering it, a segment sell-side coverage places years out.

The incumbent's switching moat is thin — which amplifies (1). Only ~11% of operators cite switching cost as the reason they stay; the largest group is actively evaluating a challenger.

Challenger momentum (Element, Ultima) is an insider signal, not yet a buyer signal. Instrument-industry respondents rate Element +0.50; prospective buyers only +0.09.

Clinical demand is an assay-class story. Blood-based early detection (MCED) registers as wanted (+0.42), but near-term clinical pull concentrates in the established assays — minimal residual disease and ctDNA monitoring lead the category.

Methodology. ReaDepth polls a standing ~360-contributor panel of sequencing & diagnostics operators, vendors and researchers each quarter. Not everyone responds in a given wave: momentum and sentiment figures are based on the contributors who responded, while consideration and interest figures reflect the standing panel's declared vendor interests. "Momentum" = net share rating a company's 12-month direction up minus down (–1 to +1); it measures sentiment and direction, not units or revenue. Figures are reported as shares of the relevant panel segment; smaller or emerging segments are flagged **DIRECTIONAL**. Wave 1 is a baseline (no trend claims). Aggregate and de-identified. **Every figure is aggregate panel opinion and directional sentiment — not a statement of fact, a valuation, or investment advice.**

01 · MARKET ENTRY

Roche/Axelios: latent demand is running ahead of the published view

Roche re-entered high-throughput sequencing in June 2026 with Axelios 1 (Sequencing-by-Expansion; a ~\$150 genome at a ~4-hour run time with integrated library prep). The prevailing external view is measured: representative sell-side coverage frames the launch as specification-competitive but a slow

clinical threat — on the order of ~100 instruments in year one, research/single-cell first, and years before it pressures Illumina's clinical consumables franchise.

The operator panel points to a materially larger latent demand pool than a ~100-unit, research-first framing implies — and, critically, it already reaches the clinical segment the external view parks in the out-years. Across the standing panel, 40% of operators list Roche among their active vendor interests, led by academic medical centers and clinical test vendors (Exhibit 1).

EXHIBIT 1 · ROCHE/AXELIOS CONSIDERATION ACROSS THE OPERATOR BASE

Academic medical centers	56%
Clinical test vendors	45%
Service labs	44%
Academic research labs	38%
Biopharma R&D	29%

Share of each operator segment listing Roche among their declared vendor interests (standing panel). Operator/buyer segments only (excludes investors, analysts and instrument-vendor staff, who do not run sequencers). ReaDepth panel of ~360 contributors; interest ≠ committed purchase.

Two features distinguish this from ordinary launch buzz. First, clinical interest is already present — 45% of clinical test vendors and 56% of academic medical centers list Roche among their interests — not deferred to a future IVD transition. Second, sentiment shows no gap between how the instrument industry and how prospective buyers rate Roche (+0.79 vs +0.79), indicating broadly-held rather than supplier-driven enthusiasm (contrast §3).

MARKET IMPLICATION

On the panel's read, the near-term constraint on Roche's trajectory looks more like manufacturing and ramp than demand generation. Incumbents and suppliers modelling a slow, research-confined clinical ramp could be under-provisioning their competitive response; in this panel the clinical evaluation cycle appears to be beginning at launch, not years after it.

02 · SWITCHING DYNAMICS

The incumbent's moat is preference, not lock-in — which amplifies Finding 1

A common assumption underpinning incumbent-durability models is that installed NGS bases are “sticky” — locked in by assay validation, reimbursement integration and revalidation cost. The panel does not support switching cost as the dominant retention mechanism. Asked directly what keeps them on their current platform, operators most often cite active evaluation of alternatives and the absence of a clearly-better challenger — not the cost of switching.

EXHIBIT 2 · WHY OPERATORS STAY ON THEIR CURRENT SEQUENCING PLATFORM

Actively evaluating / adding a challenger	50%
Genuinely satisfied with current platform	18%
No challenger clearly better yet	11%
Budget / capex freeze	11%
Switching / revalidation cost	11%

Operator-reported primary reason (single-select), as a share of respondents. Excludes “prefer not to say.”

Only ~11% name switching or revalidation cost, while the single largest group — 50% — reports already evaluating or adding a challenger. A moat built on preference and “good enough” incumbency behaves very differently from one built on structural lock-in: it is contestable by a sufficiently differentiated entrant.

MARKET IMPLICATION

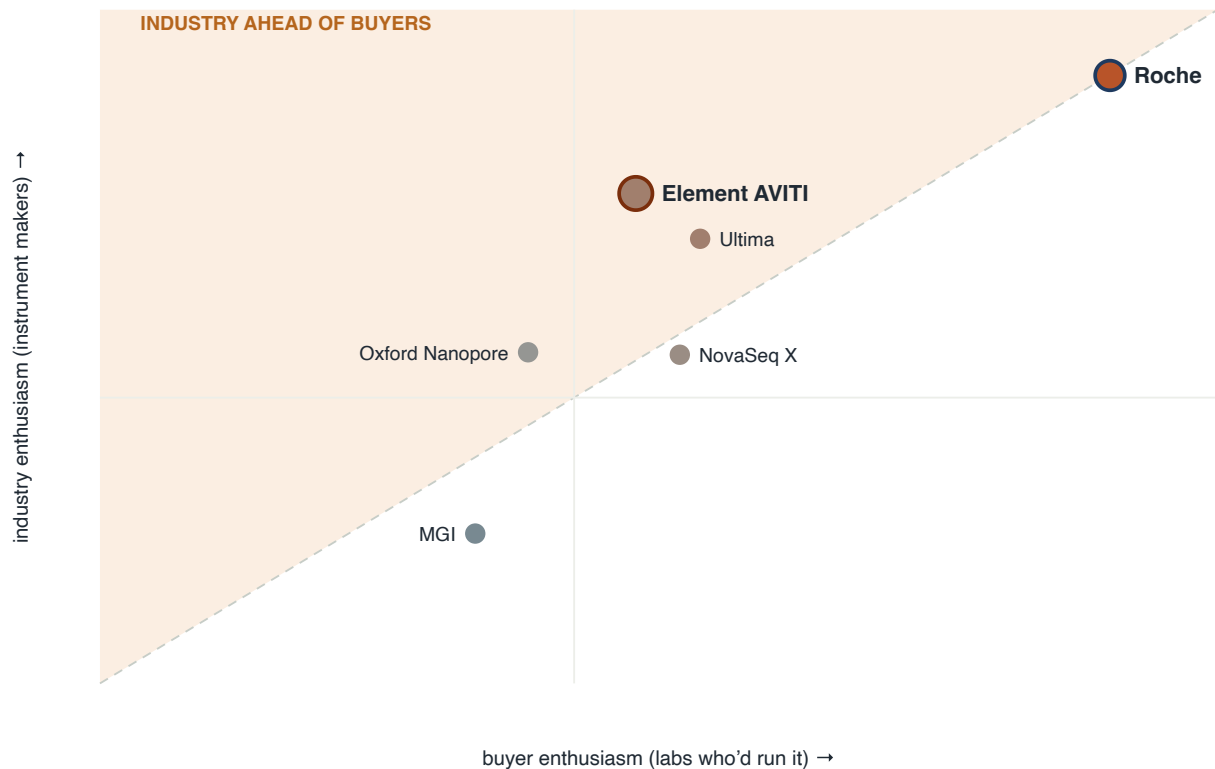
Findings 1 and 2 compound. If retention rests on preference rather than switching cost, a launch with credible specification advantages (Finding 1) could convert consideration into adoption faster than lock-in-based models predict. The durability of the installed consumables stream may warrant stress-testing against a thinner-moat assumption.

03 · CHALLENGER LANDSCAPE

Element and Ultima: industry enthusiasm running ahead of buyer adoption

Element Biosciences and Ultima Genomics are widely positioned as the credible challengers to Illumina's NovaSeq X, supported by funding and specification narratives. Decomposing panel sentiment by respondent role reveals a divergence that a blended average conceals: enthusiasm is concentrated among instrument-industry respondents, not among the operators that would purchase and run the systems.

EXHIBIT 3 · ENTHUSIASM BY RESPONDENT ROLE — INDUSTRY VS. PROSPECTIVE BUYERS



Each platform plotted by industry (instrument-maker) enthusiasm vs. buyer (laboratory) enthusiasm. Points above the diagonal indicate industry enthusiasm running ahead of buyers. Role-decomposed on a smaller sub-panel — directional.

Element — industry	+0.50	
Element — buyers	+0.09	gap +0.41
Ultima — industry	+0.39	
Ultima — buyers	+0.19	gap +0.20

Element carries a +0.41 industry-over-buyer gap (Ultima +0.20). By contrast, Roche (Finding 1) shows no such gap — its enthusiasm is shared by buyers, not just the industry. This is the distinction between a narrative the sector believes and demand the customer base has adopted.

MARKET IMPLICATION

Adoption timelines for Element and Ultima built on industry enthusiasm may run ahead of actual buyer adoption. Buyer-side enthusiasm — rather than funding, specs or industry sentiment — may be the more reliable indicator to track; on that measure the challengers' demand signal is still emerging in this panel, whereas Roche's already reads as broad-based.

04 · ASSAY DEMAND

The liquid-biopsy assay landscape: where clinical demand concentrates

Beyond the instruments, the panel rates the clinical assay categories these systems serve. Demand is not uniform. It concentrates in the established monitoring assays — minimal residual disease (MRD) and ctDNA — while blood-based multi-cancer early detection (MCED), though clearly wanted, registers as a less-mature, earlier-stage demand pool.

EXHIBIT 4 · CLINICAL ASSAY MOMENTUM

MRD	+0.88
ctDNA	+0.83
CGP	+0.53
Tumor-normal	+0.47
MCED	+0.42
Rare disease	+0.35

Net direction (growing minus contracting) by assay category, as a share of the contributors rating each. Directional for the smaller categories.

MRD and ctDNA monitoring are the clearest growth categories; comprehensive genomic profiling and tumor-normal sit in the middle; MCED is positive but trails the established assays. Read this as an assay-class signal — where near-term clinical pull is concentrating — not a verdict on any single test or provider.

MARKET IMPLICATION

For assay developers and their instrument suppliers, near-term clinical demand in the panel concentrates in MRD and ctDNA monitoring. MCED reads as a genuine but earlier-stage opportunity — category interest is real, but the pull has not yet reached the level of the established monitoring assays. The pattern suggests provisioning capacity and go-to-market against the assay class rather than the headline.

05 · INCUMBENT TRANSITION**Inside Illumina's NovaSeq 6000 → X migration**

Illumina's high-throughput base is mid-generation, moving from the NovaSeq 6000 to the newer NovaSeq X. Among the panel's Illumina operators, that shift is well underway: about 51% are already on the X or migrating now, against roughly 17% still choosing to stay on the 6000 for now.

EXHIBIT 5 · WHERE ILLUMINA OPERATORS SIT IN THE 6000 → X TRANSITION

On the X (or migrating now)	51%
Evaluating the X	14%
Staying on the 6000 (for now)	17%
Other Illumina (NextSeq, etc.)	17%

Self-reported stage, as a share of the panel's Illumina operators (excludes non-Illumina shops and no-answers).

Forward enthusiasm tracks the migration: the NovaSeq X reads +0.17 — steady, as the destination platform consolidates volume (Illumina reports the X now carries ~90% of high-throughput gigabases). The 6000 reads -0.56 — the field is actively moving off the prior generation, exactly as Illumina's

roadmap intends. Its remaining base is the shrinking tail that Roche and the challengers are targeting (\$1).

And that tail is unusually receptive. **The smaller group still on the 6000 skews sharply toward Roche** — a markedly higher share of them are already positive on Roche, or considering it, than among NovaSeq X owners. The retiring-generation base isn't just shrinking; its owners are the ones actively looking.

MARKET IMPLICATION

The generational transition itself is not the risk to Illumina — its own X is capturing the migration. The live question is the economics of the shrinking 6000 tail: it is the one pool where a differentiated entrant can take share before those runs consolidate onto the X. Watch the “staying on the 6000” cohort as the contested ground.

About this report. ReaDepth Market Intelligence is produced from a standing, quarterly-surveyed panel of ~360 sequencing and diagnostics professionals spanning instrument vendors, clinical-test vendors, service labs, biopharma R&D, and academic research and medical centers; a subset respond each wave, and these findings reflect that responding group. The panel is vendor-weighted toward the frontier-NGS names. All measures are shares and direction — not units, revenue or sample counts. Aggregate and de-identified; no individual respondent is identified. ReaDepth carries no position in any company named. All findings are aggregate opinion and directional sentiment — not statements of fact, valuations, or investment advice; forward-looking reads reflect respondents' expectations, which may prove wrong. © 2026 ReaDepth · readepth.com

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